Analysis of influence factors in variable amplitude fatigue of welded joints

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Abstract

This paper provides a framework to assess the influence of three loading spectrum parameters on the value of damage sum at failure of welded joints. A two-level full factorial design was developed by using a linear model with interaction. Experimental test results for a typical fatigue sensitive joint were used and analysed using ANOVA.

Keywords: Fatigue, Welded joints, Full factorial design

1. Introduction

Fatigue assessment of road bridges asks for consideration of welded joints. Welded joints in road bridges are subjected to variable amplitude (VA) fatigue loading. VA fatigue loads are conventionally expressed by stress range spectra (histograms of loading stress ranges).
The fatigue verification of welded joints under VA loadings is based on S-N curves, used in combination with Miner's linear damage accumulation rule [1]:

$$D = \sum_{i=1}^{n_{tot_1}} \frac{n_i}{N_i} + \sum_{j=1}^{n_{tot_2}} \frac{n_j}{N_j} \leq D_c \tag{1}$$

where:

- n_i is the number of stress range cycles corresponding to the applied nominal stress range, S_i, with $S_i >$CAFL;

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- n_j is the number of stress range cycles corresponding to the applied nominal stress range, S_j, with $S_j \leq$CAFL;
- CAFL is the fatigue limit of the S-N curve (see Figure 1);
- N_i, N_j are the number of cycles to failure corresponding to respectively S_i, S_j, according to the S-N curve;
- D_c is the critical value of the damage sum.

At present, the influence of loading spectrum parameters on the value of damage sum at failure is not clear and it is still under debate [2] [3] [4].
In this work, the influence of following loading spectrum parameters on the value of damage sum at failure, D, was studied, by developing a two-level full factorial design:

1. The minimum stress range, S_{min};
2. The number of cycles of the highest stress range, n_m;
3. The root mean square, RMS=$\sqrt{\sum(S_i^2)/n}$.

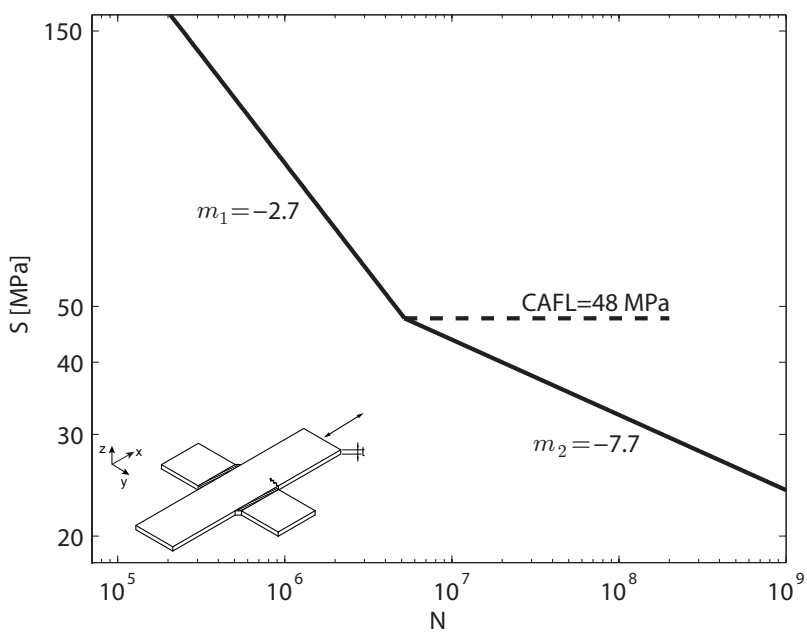

Figure 1: In-plane gusset detail: VA S-N curve

A welded in-plane gusset, which is a typical fatigue sensitive detail in road bridges, was considered (see Figure 1). The median S-N curve from the ML-MCS approach developed by D'Angelo et al. [5] (see again Figure 1) was used for computing the value of damage sum at failure, D.

2. Design of experiments

The matrix of experiments, $\underline{\underline{E}}$, of a full factorial design 2^3 is the following:

$$\underline{\underline{E}} = \begin{vmatrix} -1 & -1 & -1 \\ -1 & -1 & 1 \\ -1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} \tag{2}$$

According to the matrix of experiments, 8 VA tests were performed. Natural and coded levels of the full factorial design are represented in the Table 1. The data were analysed with a first degree linear model with interaction; the model has a constant effect, three main effects and four interaction effects:

$$Y = a_0 + a_1x_1 + a_2x_2 + a_3x_3 + a_{12}x_1x_2 + a_{13}x_1x3 + a_{23}x_2x_3 + a_{123}x_1x_2x_3 \tag{3}$$

Run	S_{min} [MPa]	n_m	RMS	X_1	X_2	X_3
1	20	100	65	-1	-1	-1
2	20	100	73	-1	-1	1
3	20	1000	65	-1	1	-1
4	20	1000	73	-1	1	1
5	40	100	65	1	-1	-1
6	40	100	73	1	-1	1
7	40	1000	65	1	1	-1
8	40	1000	73	1	1	1

Table 1: Factorial runs

The matrix of the model for the factorial design, $\underline{\underline{X}}$, is the following:

$$\underline{\underline{X}} = \begin{vmatrix} 1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 & -1 & 1 & -1 & -1 \\ 1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{vmatrix} \tag{4}$$

3. Variable amplitude fatigue tests

Eight in-plane welded gussets were tested under eight different stress range spectra (see Figure 2). Loading spectra were made of five stress range blocks, which had constant relative fatigue damage according to a one-slope S-N curve. For each test, the number of cycles at failure, N_f, was read and transformed in damage sum, D, according to Equation 1. The values of N_f and D, for each test, are shown in Table 2.

4. Results

According to damage values shown in Table 2, the vector of results is the following:

$$\underline{R} = (0.51, 0.59, 0.72, 0.52, 1.43, 1.41, 0.51, 0.86) \tag{5}$$

Run	Spectrum	N_f	D
1	A	$1.59 \cdot 10^7$	0.51
2	B	$1.66 \cdot 10^7$	0.59
3	C	$2.26 \cdot 10^7$	0.72
4	D	$1.46 \cdot 10^7$	0.52
5	E	$6.43 \cdot 10^6$	1.43
6	F	$6.01 \cdot 10^6$	1.41
7	G	$2.29 \cdot 10^6 6$	0.51
8	H	$3.68 \cdot 10^6$	0.86

Table 2: Fatigue test results

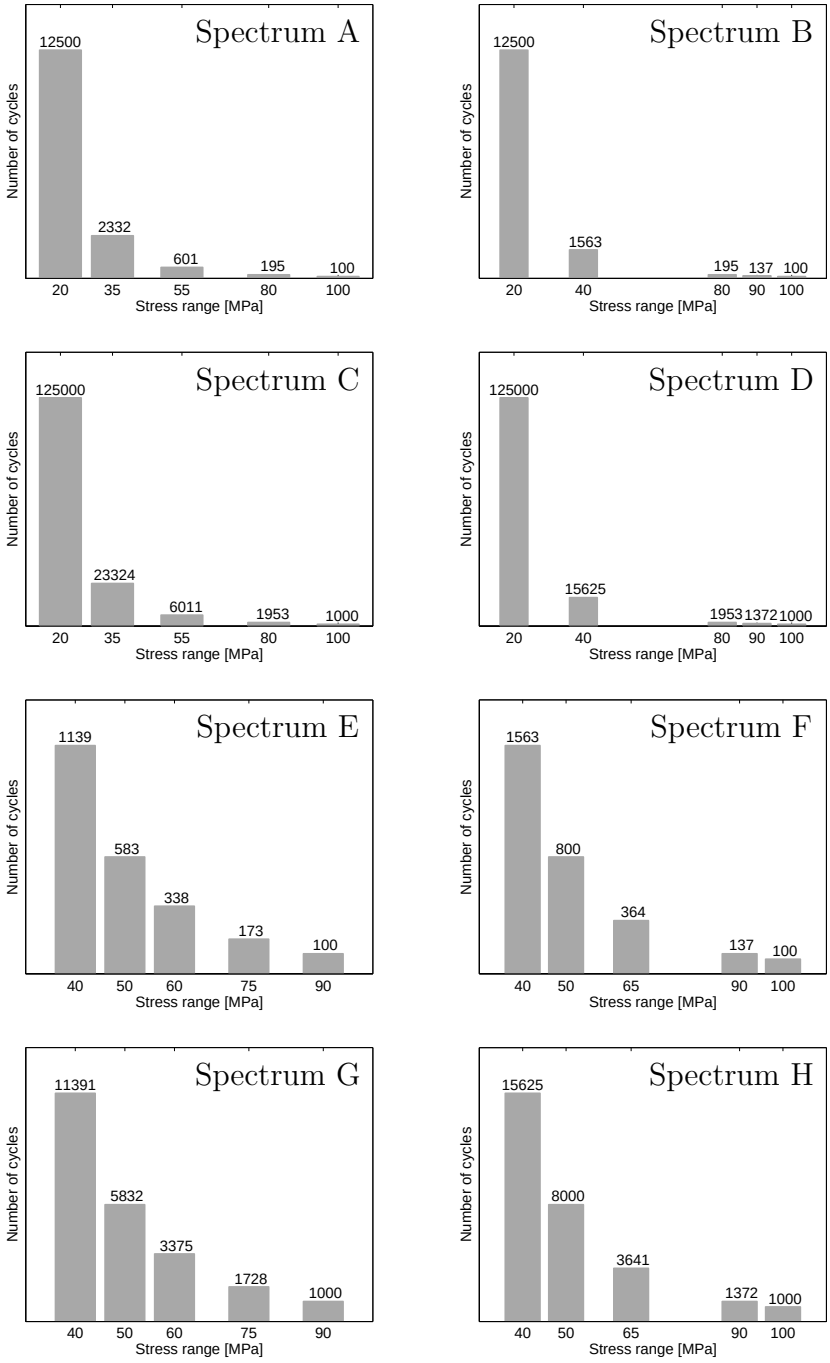

Figure 2: Loading spectra

	Main effects				Interaction effects			
Coefficient	a_0	a_1	a_2	a_3	a_{12}	a_{13}	a_{23}	a_{123}
Half effect	0.818	0.235	-0.166	0.025	-0.202	0.057	0.011	0.083
Rel. half effect		28.8%	-20.3%	3.1%	-24.7%	6.9%	1.3%	10.1%

Table 3: Factorial runs

Effect	SS	DF	MS	F	p
a_0	5.35	1			
a_1	0.44	1	0.44	41.6	0.8%
a_2	0.22	1	0.22	20.7	2.0%
a_{12}	0.33	1	0.33	30.7	1.2%
a_{123}	0.05	1	0.05	5.1	10.8%
residual	0	3	0.01		
	6.42	8			

Table 4: ANOVA table

The coefficients of the linear model (see Equation 3), were computed using a least square (LS) fit. Half effects and relative half effects are shown in Table 3. The importance of relative half effects is shown in Figure, where an arbitrary threshold has been set at 10%. The effects a_3, a_{13} and a_{23} were disqualified and the ANOVA was performed (see Table 4); the coefficient of determination for this model is R^2=99.50%. The normal plot of the effects (see Figure 4) confirms disqualification of a_3, a_{13} and a_{23}, since these effects are aligned on a straight line.
The ANOVA table (see Table TABanova) gives following information:

- The effect of the minimum stress range is significant because there is only 0.8% probability for the effect to be random;
- The effect of the highest stress range block length is significant because there is only 2.0% probability for the effect to be random;
- The interaction effect between S_{min} and n_m is significant because there is only 1.2% probability for the effect to be random;
- The three term-interaction effect has 10.8% probability to be random. This is supported also from the normal plot of effects (see Figure 4), since a_{123} is aligned on a straight line.

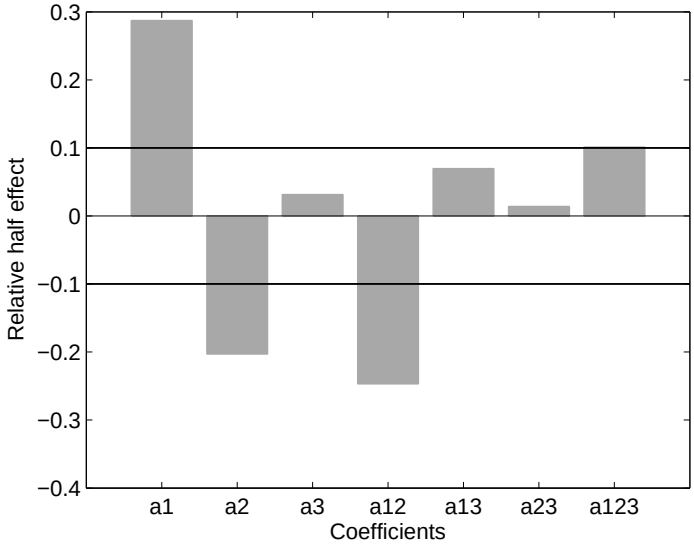

Figure 3: Relative half effects

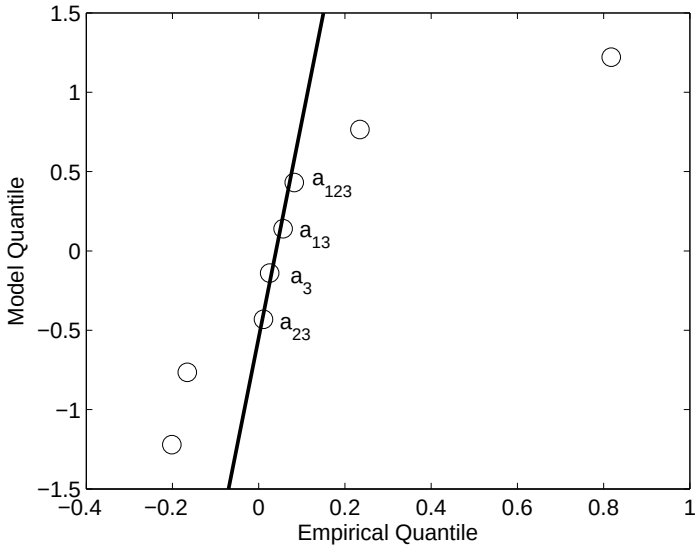

Figure 4: Normal plot of the effects

5. Conclusions

The two-level full factorial design developed in this study, allowed for the identification of the fatigue damage linear model with interactions. The analysis of data further provided information on the influence of three loading spectrum parameters (minimum stress range, S_{min}, block length of maximum stress range, n_m, and root mean square of spectrum, RMS) on the value of damage sum at failure, for a typical fatigue sensitive welded joint.
The analysis of relative effects and the normal plot of effects were used to reject the RMS main effect and the two interaction effects RMS-S_{min}, RMS-n_m. The coefficient of determination of the final model is R^2=99.50%.
Significance of effects of the final model was assessed using ANOVA: the most significant effect was found to be the effect S_{min}, which is related to the position of the loading spectrum with respect to the CAFL. The main effect n_m and the interaction effect S_{min}-n_m were found to be highly significant while the three-factor interaction effect was found to be not clearly distinguishable from random noise.

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